

PYTHON APPLICATION INNOVATIONS & INSTALLATION GUIDE

This document provides installation instructions, database configuration guidance, and an overview of the additional functionalities and innovations implemented in the Python application.

1. PYTHON INSTALLATION

Download and install the latest version of Python for Windows from:

<https://www.python.org/downloads/>

During installation, ensure that:

- “Add Python to PATH” is enabled.
- The installation is completed using the default settings.

2. REQUIRED PYTHON PACKAGES

Open Command Prompt or Terminal and run the following commands individually:

1. **Install MySQL Connector**
`python -m pip install mysql-connector-python`
2. **Install Neo4j Driver**
`python -m pip install neo4j`
3. **Install NetworkX**
`python -m pip install networkx`
4. **Install Matplotlib**
`python -m pip install matplotlib`

3. VERIFY THE INSTALLATION

Run the following command to verify the installation:

```
python -version
```

4. DATABASE CONFIGURATION

```
mysql = {
    "host": "127.0.0.1",
    "port": 3306,
    "user": "root",
    "password": "your_password",
    "database": "appdbproj",
    "connection_timeout": 5,
    "use_pure": True
}

NEO4J_URI = "bolt://localhost:7687"
NEO4J_USER = "your_neo4j_user_name"
NEO4J_PASSWORD = "your_password"
NEO4J_DATABASE = "appdbprojneo4j"
```

5. INNOVATIONS AND ADDITIONAL FUNCTIONALITIES

TKINTER GRAPHICAL USER INTERFACE (GUI)

- Replaced console-based input() and print() operations with a user-friendly Tkinter interface.
- Implemented dedicated popup windows for each functionality.
- Added input validation messages and structured result display areas.
- Designed the application to run in a graphical desktop environment such as an Azure VM connected via Remote Desktop (RDP).

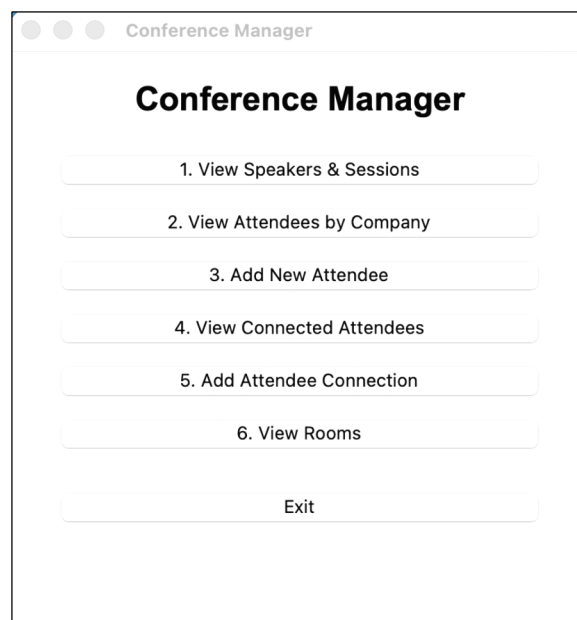


Figure 1. Tkinter GUI

TKINTER TREEVIEW TABLES

- Replaced plain text outputs with structured Tkinter Treeview tables.
- Improved readability and data presentation for database query results.

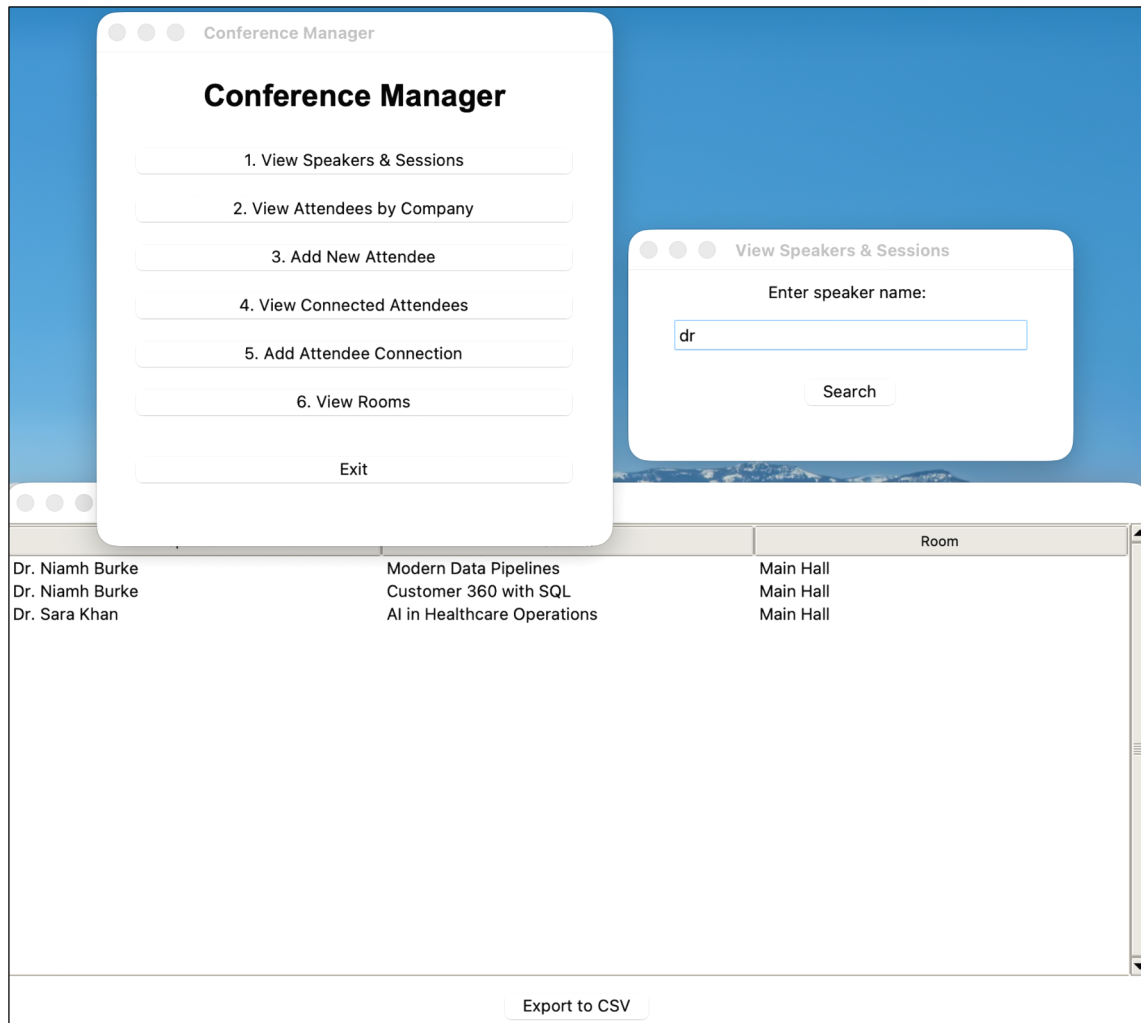


Figure 2. Tkinter Treeview tables

NEO4J GRAPH VISUALIZATION

- Integrated NetworkX and Matplotlib to visualize Neo4j database relationships.
- Automatically generates node and edge graphs based on attendee connections.
- Customized graph and window titles to improve usability and clarity.

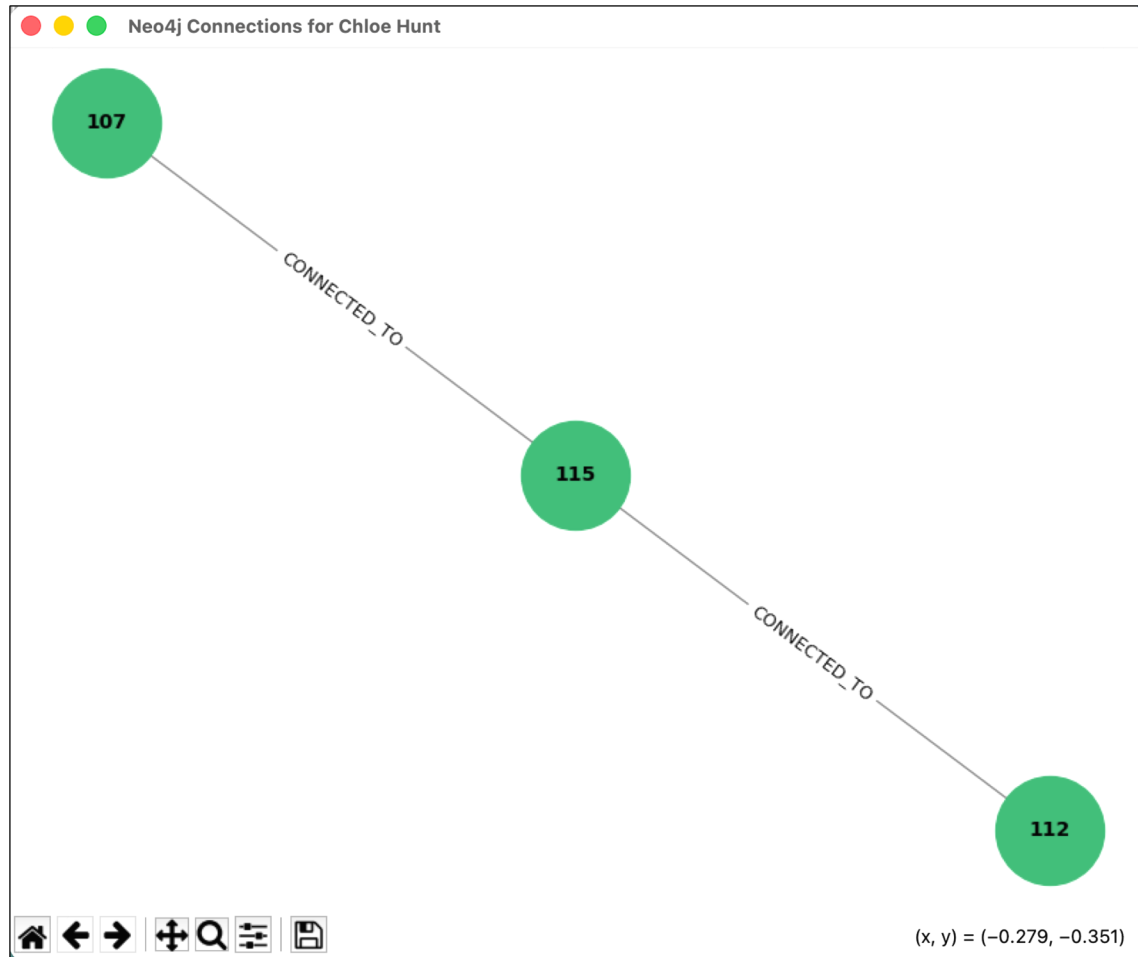


Figure 3. Graph visualization of Neo4j connections

CSV EXPORT FUNCTIONALITY

- Added one-click CSV export support for all table views.
- Allows users to select a custom save location.
- Exports data together with column headers for improved usability.

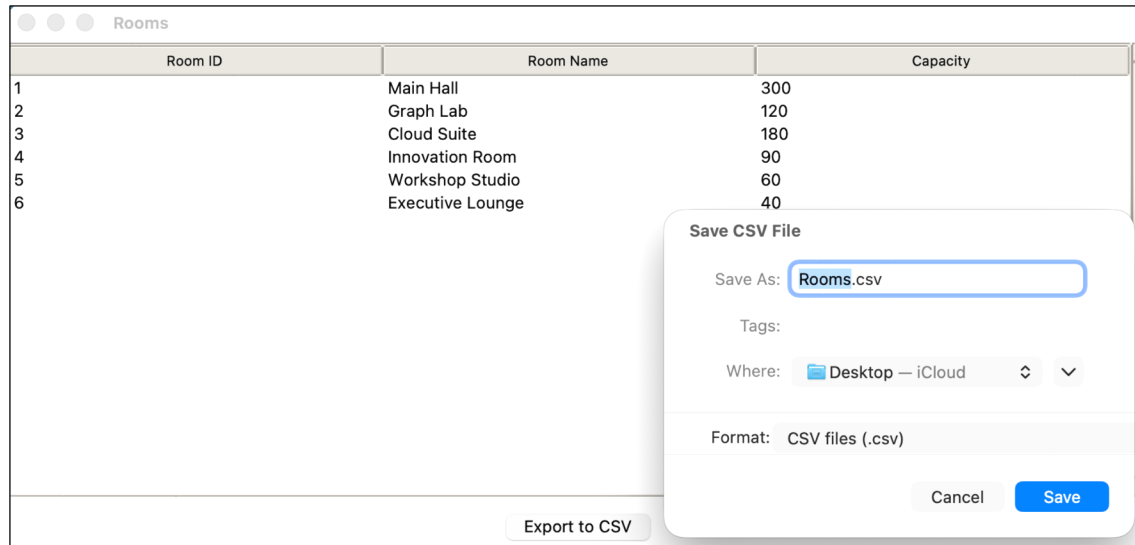


Figure 4. CSV Export

DATABASE ERROR HANDLING

- Implemented exception handling for MySQL and Neo4j database connections.
- Prevents the application from appearing unresponsive when connection failures occur.
- Provides clear feedback messages to the user in case of errors.

6. SUMMARY

The implemented enhancements significantly improve the usability, reliability, and professional quality of the application. The introduction of a graphical user interface, structured tables, graph visualizations, CSV export functionality, and robust error handling demonstrates the practical application of modern Python development techniques while improving the overall user experience.